WORKED SOLUTION

STEP II 2013, Question 2

Pure • Reduction Formulae

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s2-q2

The key idea

The integrand $x^n(1-x)^n$ is symmetric under the reflection $x \mapsto 1-x$, and that symmetry lets us swap the roles of x and $1-x$ for free. Integration by parts then trades a power of x for a power of $1-x$, and the identity $x+(1-x)=1$ lets us split a product back into integrals we already know. Together these give a reduction formula, which unwinds to factorials for whole-number n and also works for half-integers.

Solution

Part (i)

Write $J_n = \int_0^1 x^{n-1}(1-x)^n \,\mathrm{d}x$ for $n \geq 1$. Substitute $u = 1-x$, so $\mathrm{d}x = -\mathrm{d}u$ and the limits $0, 1$ become $1, 0$:

$$J_n = \int_1^0 (1-u)^{n-1}u^n\,(-\mathrm{d}u) = \int_0^1 u^n(1-u)^{n-1}\,\mathrm{d}u,$$

which is the required equality (the name of the dummy variable doesn't matter).

Adding the two equal forms of J_n,

$$\begin{aligned} 2J_n &= \int_0^1 \left[x^{n-1}(1-x)^n + x^n(1-x)^{n-1}\right]\mathrm{d}x \\ &= \int_0^1 x^{n-1}(1-x)^{n-1}[(1-x)+x]\mathrm{d}x = I_{n-1}. \end{aligned}$$

Next integrate I_n by parts, with $u = x^n$ and $\mathrm{d}v = (1-x)^n\,\mathrm{d}x$, so $v = -\dfrac{(1-x)^{n+1}}{n+1}$:

$$I_n = \left[-\frac{x^n(1-x)^{n+1}}{n+1}\right]_0^1 + \frac{n}{n+1}\int_0^1 x^{n-1}(1-x)^{n+1}\,\mathrm{d}x.$$

The boundary term vanishes at both ends: at $x=1$ because of $(1-x)^{n+1}$, and at $x=0$

because of x^n with $n \geq 1$. This gives the stated formula

$$I_n = \frac{n}{n+1} \int_0^1 x^{n-1}(1-x)^{n+1}\,\mathrm{d}x.$$

Ask yourself How do we get from this integral back to I_n and I_{n-1}?

Split off one factor of $1-x$ and write it as $1-x$ literally:

$$x^{n-1}(1-x)^{n+1} = x^{n-1}(1-x)^n\,(1-x) = x^{n-1}(1-x)^n - x^n(1-x)^n.$$

Integrating, the right-hand side gives $J_n - I_n = \frac{1}{2}I_{n-1} - I_n$, using the result above. So

$$I_n = \frac{n}{n+1}\left(\tfrac{1}{2}I_{n-1} - I_n\right) \quad \Longrightarrow \quad I_n \cdot \frac{2n+1}{n+1} = \frac{n}{2(n+1)}I_{n-1},$$

and multiplying by $\dfrac{n+1}{2n+1}$,

$$I_n = \frac{n}{2(2n+1)}\,I_{n-1},$$

as required. Notice that nothing in this argument used n being an integer: only $n \geq 1$ was needed (for the boundary term and so that every integrand is continuous on $[0,1]$). We shall use that in part (iii).

Part (ii)

We prove $I_n = \dfrac{(n!)^2}{(2n+1)!}$ by induction, starting from $n=0$: $I_0 = \int_0^1 1\,\mathrm{d}x = 1 = \dfrac{(0!)^2}{1!}$. Suppose the formula holds for $n-1$, where $n \geq 1$. Then by part (i)

$$\begin{aligned} I_n = \frac{n}{2(2n+1)} \cdot \frac{((n-1)!)^2}{(2n-1)!} &= \frac{n^2((n-1)!)^2}{2n(2n+1)(2n-1)!} \\ &= \frac{(n!)^2}{(2n+1)!}, \end{aligned}$$

where we multiplied top and bottom by n and used $2n(2n+1)(2n-1)! = (2n+1)!$. This completes the induction, so the formula holds for every positive integer n (and for $n=0$).

Part (iii)

With $x = \sin^2\theta$ we have $\mathrm{d}x = 2\sin\theta\cos\theta\,\mathrm{d}\theta$, and θ runs from 0 to $\frac{\pi}{2}$. On this interval $\sin\theta$ and $\cos\theta$ are non-negative, so

$$(x(1-x))^{1/2} = (\sin^2\theta\cos^2\theta)^{1/2} = \sin\theta\cos\theta.$$

Ask yourself Why do we need $\sin\theta\cos\theta \geq 0$ here?

Because $\sqrt{t^2} = |t|$, not t; on $[0, \frac{\pi}{2}]$ the modulus can simply be dropped. Hence

$$I_{\frac{1}{2}} = \int_0^{\pi/2} \sin\theta\cos\theta \cdot 2\sin\theta\cos\theta \,\mathrm{d}\theta = \int_0^{\pi/2} \tfrac{1}{2}\sin^2 2\theta \,\mathrm{d}\theta$$
$$= \int_0^{\pi/2} \tfrac{1}{4}(1-\cos 4\theta)\mathrm{d}\theta = \tfrac{1}{4}\Big[\theta - \tfrac{1}{4}\sin 4\theta\Big]_0^{\pi/2} = \frac{\pi}{8}.$$

For $I_{\frac{3}{2}}$, use the reduction formula from part (i) with $n = \frac{3}{2}$, which is allowed since $\frac{3}{2} \geq 1$ and the derivation never needed n to be an integer:

$$I_{\frac{3}{2}} = \frac{3/2}{2\cdot 4} I_{\frac{1}{2}} = \frac{3}{16}\cdot\frac{\pi}{8} = \frac{3\pi}{128}.$$

As a cross-check, the same substitution gives directly

$$I_{\frac{3}{2}} = \int_0^{\pi/2} 2\sin^4\theta\cos^4\theta \,\mathrm{d}\theta = \frac{1}{8}\int_0^{\pi/2} \sin^4 2\theta \,\mathrm{d}\theta = \frac{1}{8}\cdot\frac{3\pi}{16} = \frac{3\pi}{128},$$

using $\sin^4\phi = \frac{3}{8} - \frac{1}{2}\cos 2\phi + \frac{1}{8}\cos 4\phi$.

Checks

- $I_1 = \int_0^1 (x - x^2)\,\mathrm{d}x = \frac{1}{6}$, and the formula gives $\frac{1}{3!} = \frac{1}{6}$. Likewise $I_2 = \int_0^1 (x^2 - 2x^3 + x^4)\,\mathrm{d}x = \frac{1}{3} - \frac{1}{2} + \frac{1}{5} = \frac{1}{30} = \frac{4}{5!}$.
- $I_{\frac{1}{2}}$ is the area under the semicircle $y = \sqrt{\frac{1}{4} - (x - \frac{1}{2})^2}$ of radius $\frac{1}{2}$, which is $\frac{1}{2}\pi\big(\frac{1}{2}\big)^2 = \frac{\pi}{8}$.
- Numerically, $I_{\frac{3}{2}} \approx 0.073631$, matching $3\pi/128$; the reduction formula was also confirmed by computer for non-integer n such as $n = 2.7$.

Common slips

- Forgetting to check that the boundary term in the integration by parts vanishes, especially at $x = 0$.
- Writing $\sqrt{\sin^2\theta\cos^2\theta} = \sin\theta\cos\theta$ without noting the sign on $[0, \frac{\pi}{2}]$.
- Computing $I_{\frac{3}{2}}$ from the factorial formula of part (ii), which is only proved for integers; the reduction formula is the safe route.

Beyond STEP

I_n is a special case of the beta function $B(p,q) = \int_0^1 x^{p-1}(1-x)^{q-1}\,\mathrm{d}x$, here $I_n = B(n+1, n+1)$, and part (ii) is the integer case of $B(p,q) = \Gamma(p)\Gamma(q)/\Gamma(p+q)$. Putting

$n = \frac{1}{2}$ into that identity recovers $\Gamma(\frac{1}{2}) = \sqrt{\pi}$, which is the deeper reason a π appears in part (iii). Integrals like these lie behind the beta distribution in statistics and Wallis's product for π; Siklos, *Advanced Problems in Mathematics*, has more reduction-formula problems in this spirit.

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